ELEG 5693 Assignment # 2

1. The pdf of the exponential RV is $f(x) = \frac{1}{u}\exp\left(-\frac{x}{u}\right)$, for $x > 0$, and $f(x) = 0$ otherwise. Find the mean and variance of X.

2. Find the mean of the RV X with pdf $f(x) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left\{-\frac{(x-m)^2}{2\sigma^2}\right\}$. Show the procedures.

3. The joint pdf of (X, Y) is $f(x, y) = \exp(-y)$, for $0 < x < y$, and $f(x, y) = 0$ otherwise.

   (a) Find $f(x)$

   (b) Find $f(y)$

   (c) Are they indepdent?

   (d) Are they correlated?

   (e) Find $f(x|y)$.

4. X and Y are two independent Gaussian RVs with mean μ and variance σ^2, *i.e.*, $X \sim N(\mu, \sigma^2)$, $Y \sim N(\mu, \sigma^2)$. Find the correlation between $Z_1 = \alpha X + \beta Y$ and $Z_2 = \alpha X - \beta Y$.

5. If 20Mhz of total spectrum is allocated for a FDD wireless cellular system and each simplex channel has 25KHz bandwidth, find:

   (a) the number of duplex channels

   (b) the total number of channels per cell site, if N = 12 cell reuse is used.

   (c) what is the reuse factor?